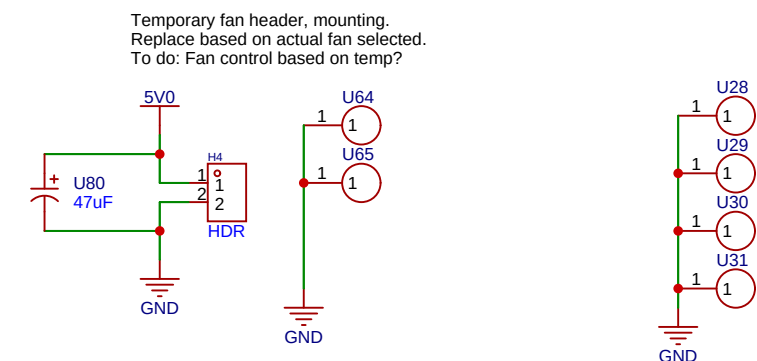


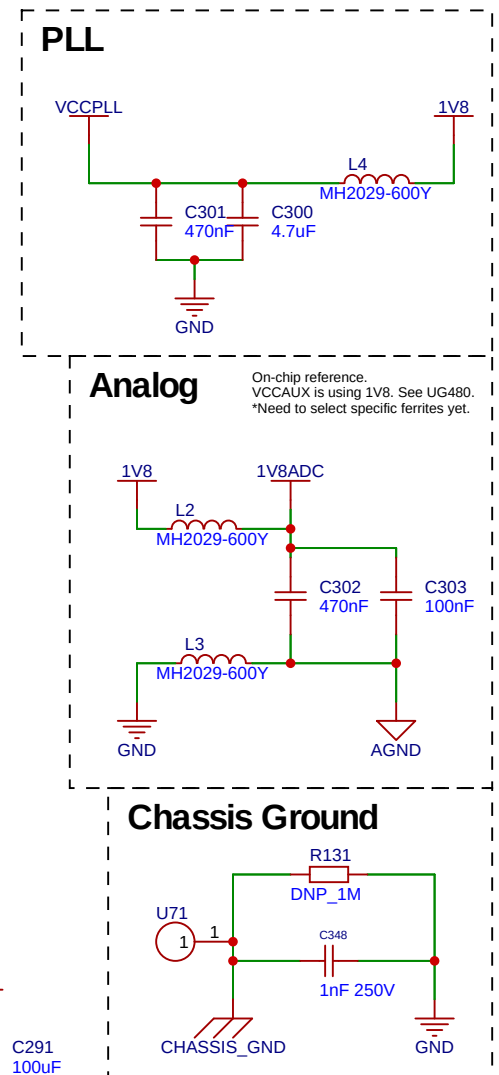
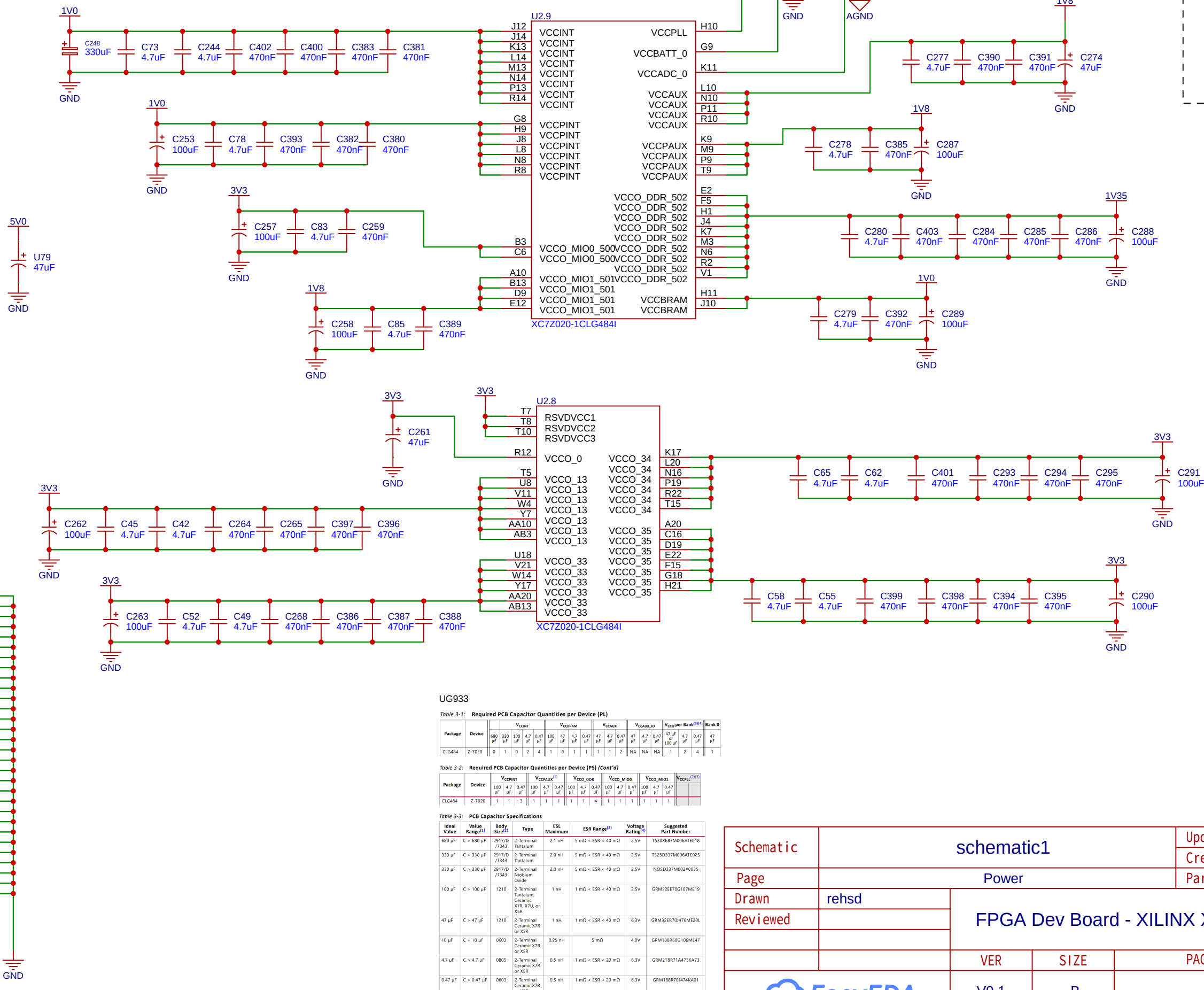
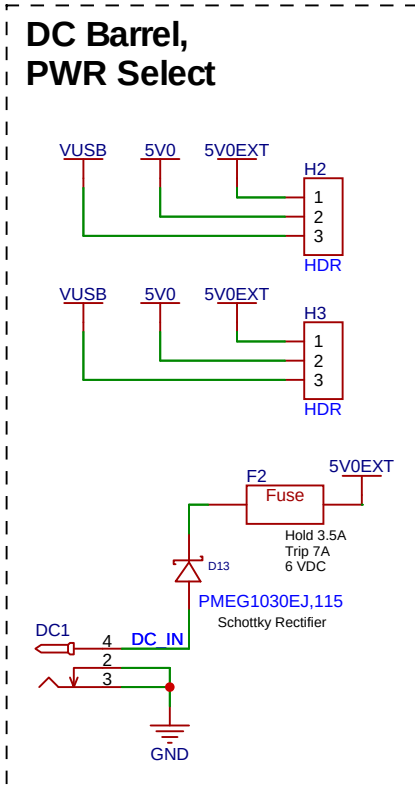
Future Updates :

- power off sequencing
- 0.675 power indicator design (different mosfet for lower voltage?)
- Programming header HS3 should be 2.0mm instead of 2.54mm
- Move R267 from DDR to closer to SoC
- Move C193 closer to U61
- Incorrect part # on 33MHz oscillator - installed 25MHz for now
- Additional oscillator for FPGA fabric (without Eth IC)?
- Power header for 3.3V
- Onboard LED for PL
- Onboard buttons for PS and PL
- Power rail selection jumpers don't seem to be needed. Always leave set to external DC barrel power.
- Improve Ethernet reset circuit?
- Add in HDMI controller (ADV7513?)
- Power coming in from HDMI connection - something needed to prevent this when the PCB is off?
- Add RTC - example: https://www.xilinx.com/support/documents/boards_and_kits/zynq-7000/zc706-schematic-xtp215-rev1-1.pdf
- On inductor pads, shrink .9mm vias to .6 so that they are filled



Schematic	schematic1			Update Date	2026-05-10
				Create Date	2026-05-10
Page	Misc			Part Number	
Drawn	rehsd	FPGA Dev Board - XILINX XC7Z020-1CLG484I v0.07			
Reviewed					
		VER	SIZE	PAGE	1 OF 13
		V0.1	B		

...



UG933

Table 3-1: Required PCB Capacitor Quantities per Device (PL)

Package	Device	V _{CCINT}						V _{CCSRAM}						V _{CCAUX}			V _{CCAUX_IO}			V _{CCO} per Bank ^(30A)				Bank C
		680 μ F	330 μ F	100 μ F	4.7 μ F	0.47 μ F	100 μ F	4.7 μ F	0.47 μ F	4.7 μ F	0.47 μ F	4.7 μ F	0.47 μ F	4.7 μ F	0.47 μ F	4.7 μ F	0.47 μ F	4.7 μ F or 100 μ F	4.7 μ F	0.47 μ F	4.7 μ F	0.47 μ F		
CLG484	Z-7020	0	1	0	2	4	1	0	1	1	1	1	1	2	NA	NA	NA	1	2	4	1			

Table 3-2: Required PCB Capacitor Quantities per Device (PS) (Cont'd)

Package	Device	V _{CCPINT}			V _{CCPAUX} ⁽¹⁾			V _{CCO_DDR}			V _{CCO_MIO0}			V _{CCO_MIO1}			V _{CCPLL} ⁽²⁾⁽³⁾	
		100 μF	4.7 μF	0.47 μF	100 μF	4.7 μF	0.47 μF	100 μF	4.7 μF	0.47 μF	100 μF	4.7 μF	0.47 μF	100 μF	4.7 μF	0.47 μF		
CLG484	Z-7020	1	1	3	1	1	1	1	1	4	1	1	1	1	1	1		

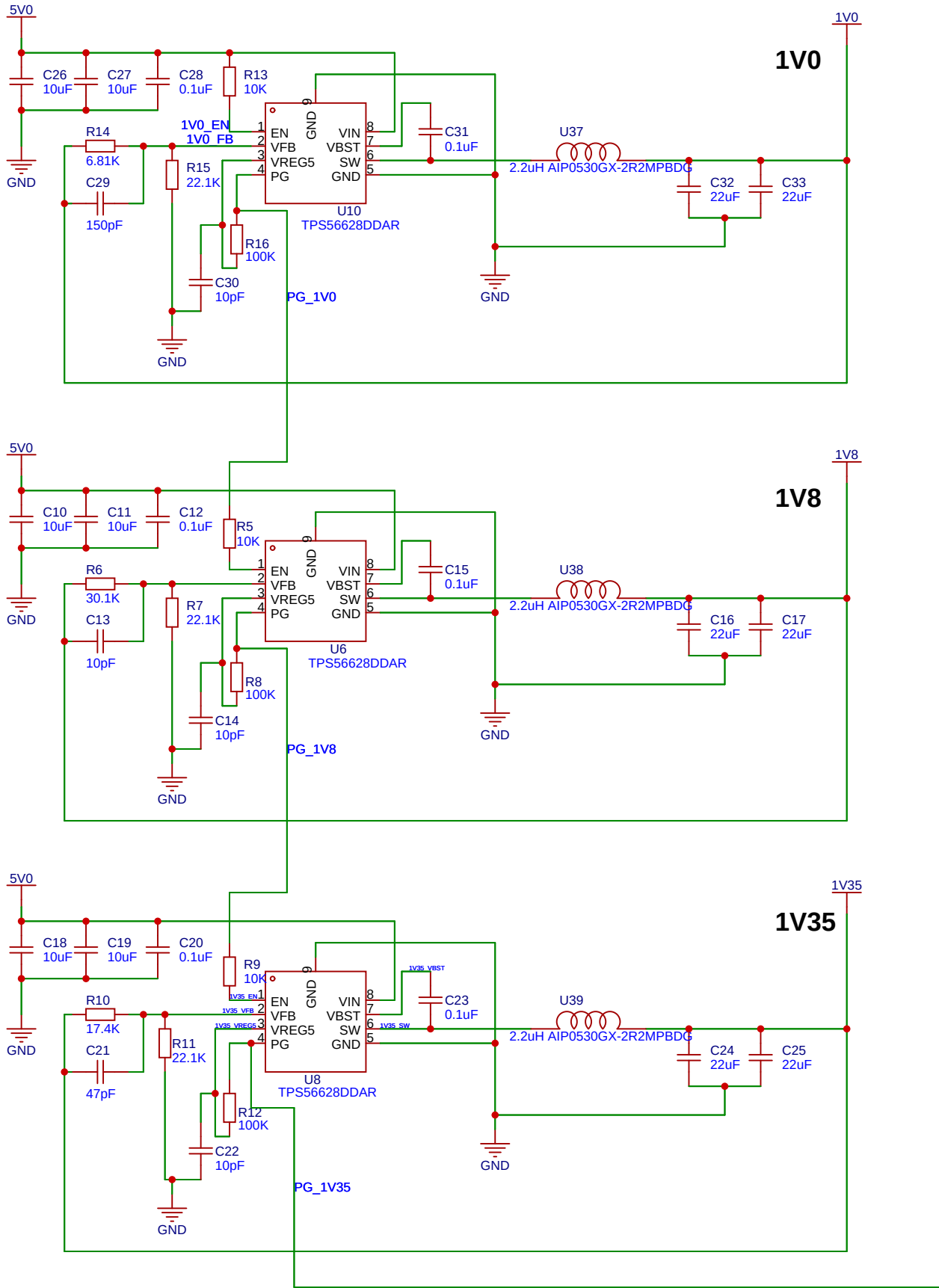
Table 3-3: PCB Capacitor Specifications

Ideal Value	Value Range ⁽¹⁾	Body Size ⁽²⁾	Type	ESL Maximum	ESR Range ⁽³⁾	Voltage Rating ⁽⁴⁾	Suggested Part Number
680 μ F	C > 680 μ F	2917/D /F343	2-Terminal Tantalum	2.1 nH	5 m Ω < ESR < 40 m Ω	2.5V	TS10X068H7M006AT018
330 μ F	C > 330 μ F	2-Terminal /F343	2-Terminal Tantalum	2.0 nH	5 m Ω < ESR < 40 m Ω	2.5V	TS2SD330H7M006AT025
330 μ F	C > 330 μ F	2-Terminal /F343	2-Terminal Niobium Oxide	2.0 nH	1 m Ω < ESR < 40 m Ω	2.5V	NOSD3337M002K0035
100 μ F	C > 100 μ F	1210	2-Terminal Tantalum, Ceramic X7R, X7U, or X5R	1 nH	1 m Ω < ESR < 40 m Ω	2.5V	GRM32E7E0G107M19
47 μ F	C > 47 μ F	1210	2-Terminal Ceramic X7R or X5R	1 nH	1 m Ω < ESR < 40 m Ω	6.3V	GRM32E7A706J20L
10 μ F	C > 10 μ F	0603	2-Terminal Ceramic X7R or X5R	0.25 nH	5 m Ω	4.0V	GRM188R60G106M47
4.7 μ F	C > 4.7 μ F	0805	2-Terminal Ceramic X7R or X5R	0.5 nH	1 m Ω < ESR < 20 m Ω	6.3V	GRM218R71A475K63
0.47 μ F	C > 0.47 μ F	0603	2-Terminal Ceramic X7R or X5R	0.5 nH	1 m Ω < ESR < 20 m Ω	6.3V	GRM188R71A474K401

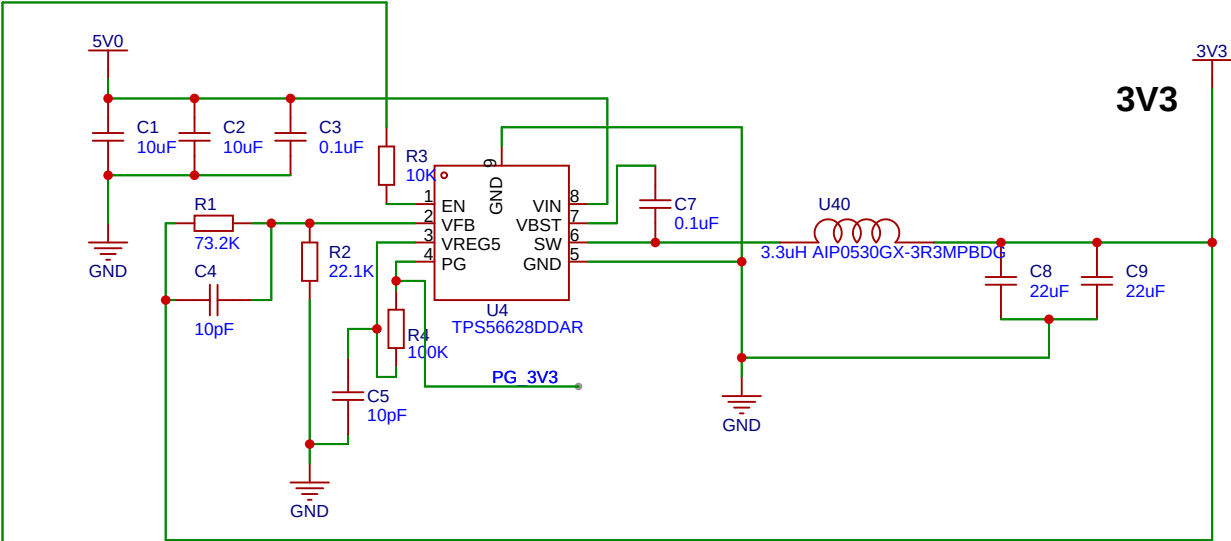
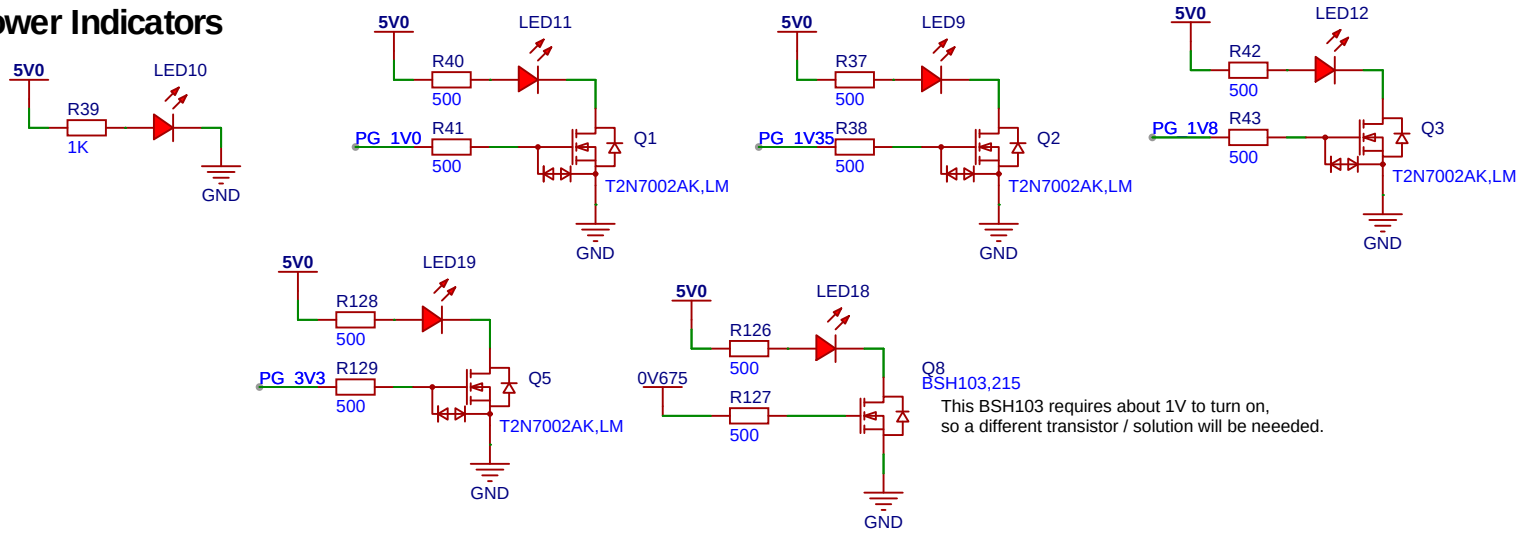
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				Create Date	2026-05-10
Page	Power			Part Number	
Drawn	rehsd	FPGA Dev Board - XILINX XC7Z020-1CLG484I v0.07			
Reviewed					
		VER	SIZE	PAGE	2 OF 13
		V0.1	B		

Buck Converters

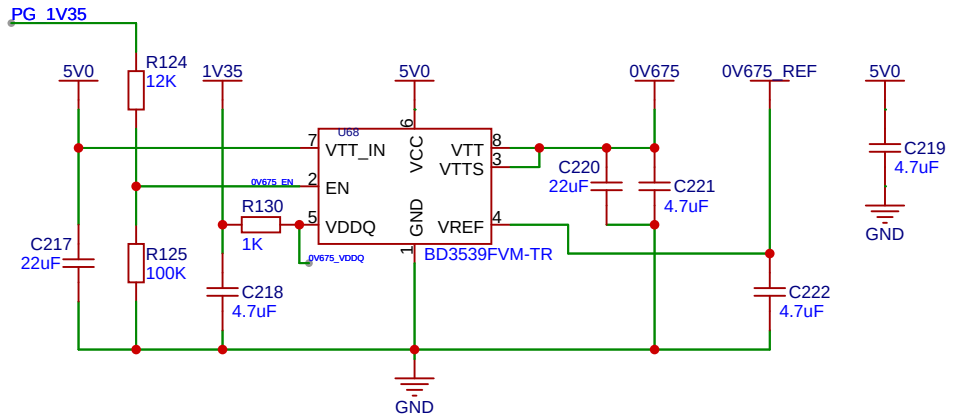
Sequence: 1V0, 1V8, 1V35, 3V3



Power Indicators



DDR3L Bus Termination Voltage (VTT)

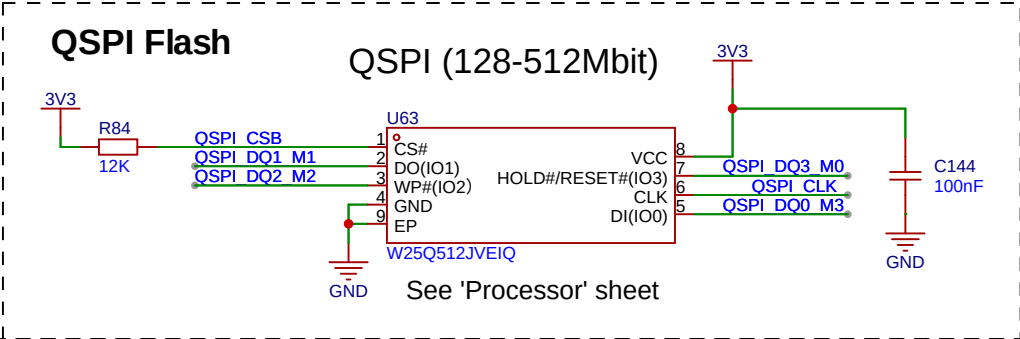


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Page	Regulators			Part Number	
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Reviewed					
		VER	SIZE	PAGE	3 OF 13
		V0.1	B	EasyEDA.com	

Programming

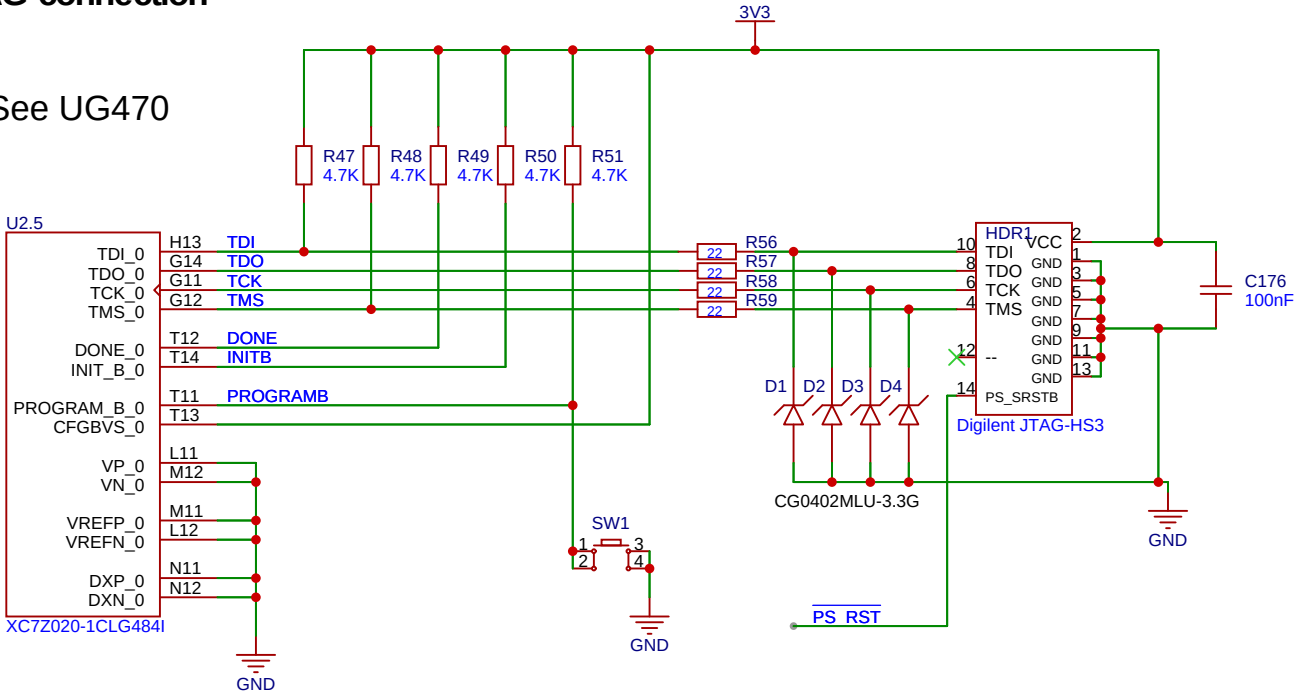
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To do:
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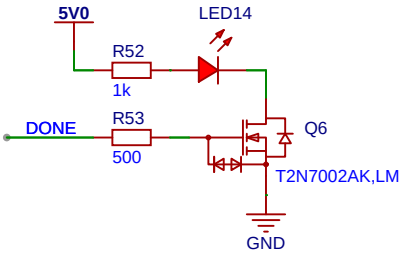


JTAG connection

See UG470



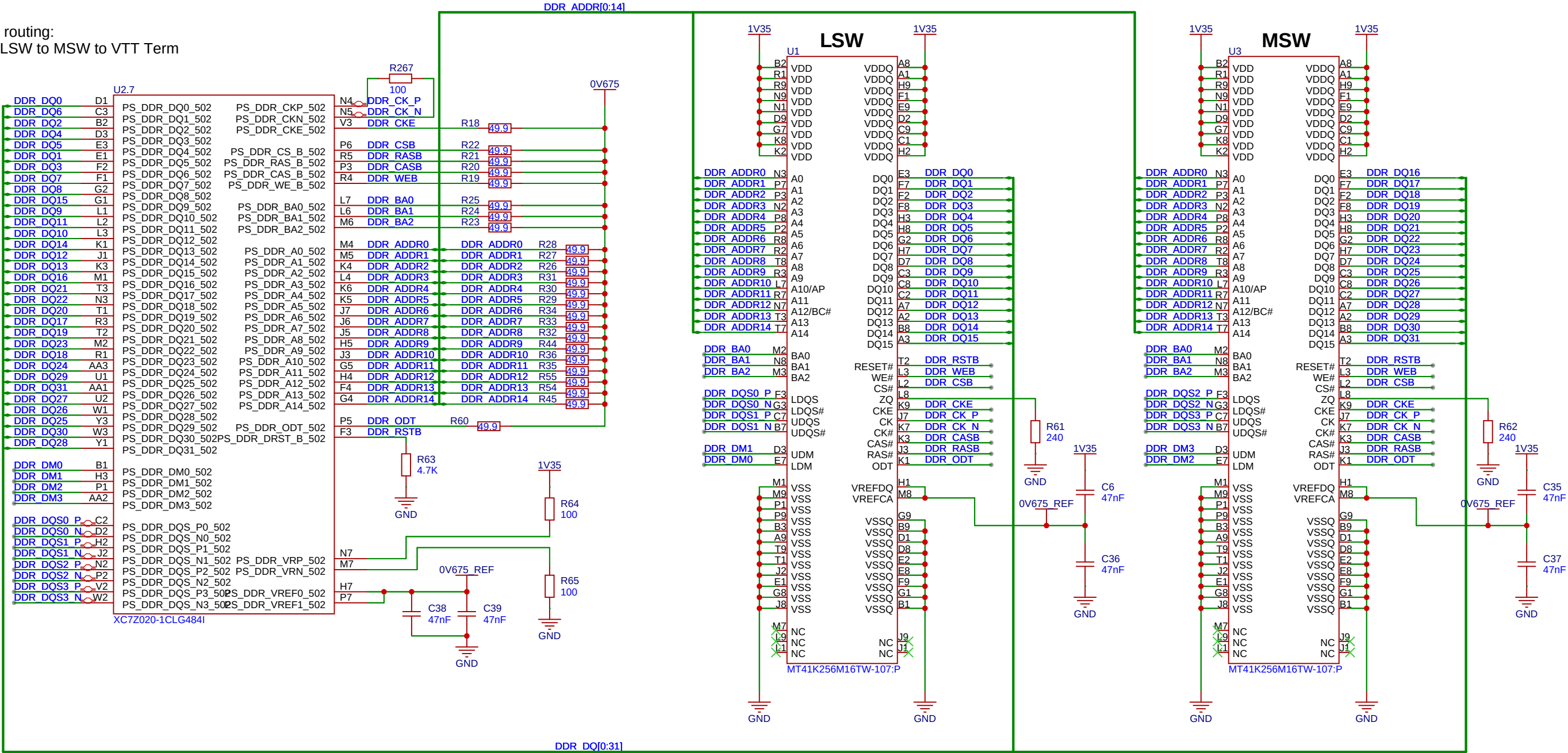
Done indicator



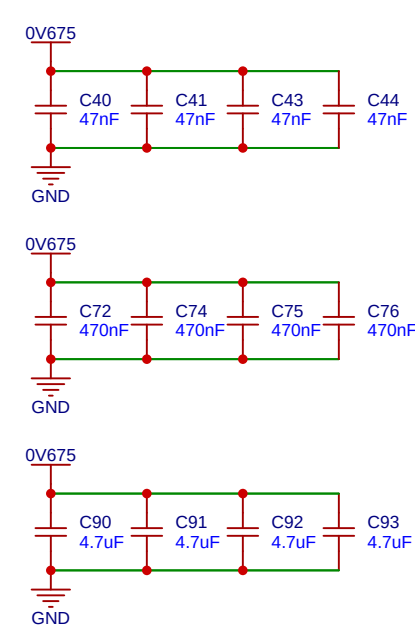
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Page	Program			Part Number	JLCPCB-002
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Reviewed	EasyEDA				
		VER	SIZE	PAGE	4 OF 13
EasyEDA		V0.1	A4	EasyEDA.com	

DDR3L

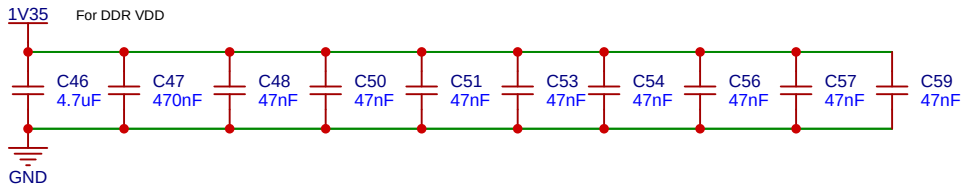
Fly-by routing:
PS to LSW to MSW to VTT Term



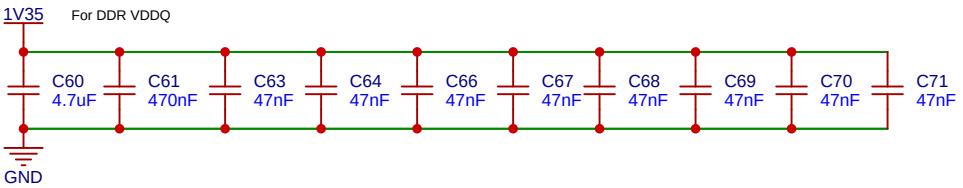
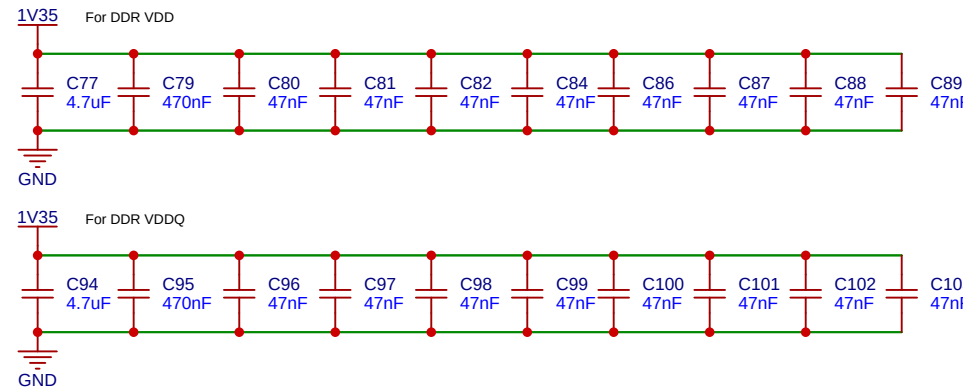
Bypass Caps DDR Term



Bypass Caps DDR MSW



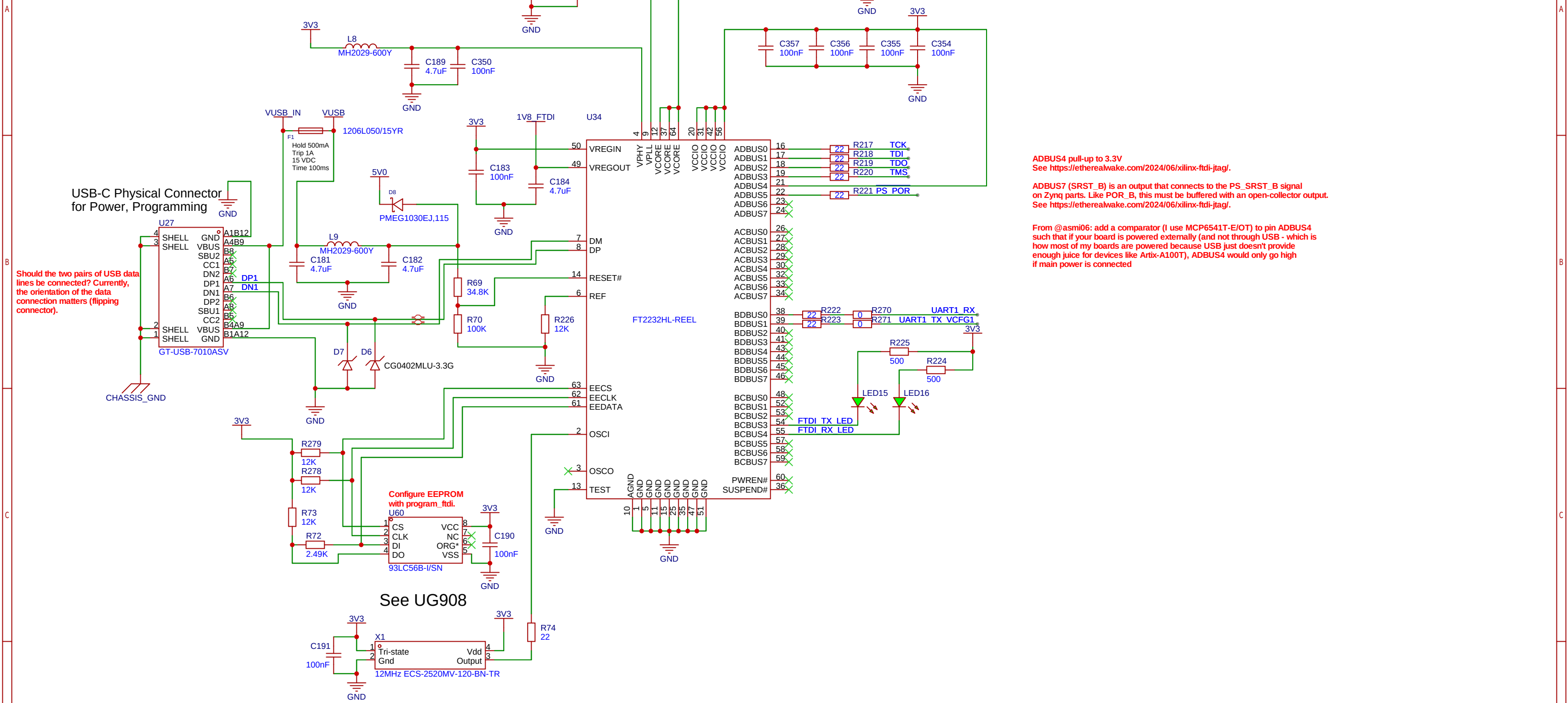
Bypass Caps DDR LSW



Schematic	schematic1			Update Date	2026-05-10			
				Create Date	2026-05-10			
Page	DDR3L			Part Number				
Drawn	rehsd	FPGA Dev Board - XILINX XC7Z020-1CLG484I v0.07						
Reviewed								
		VER	SIZE	PAGE	5	OF	13	
		V0.1	B	EasyEDA.com				

USB

Programming, Serial



ADBUS4 pull-up to 3.3V
See <https://etherealwake.com/2024/06/xilinx-ftdi-jtag/>.

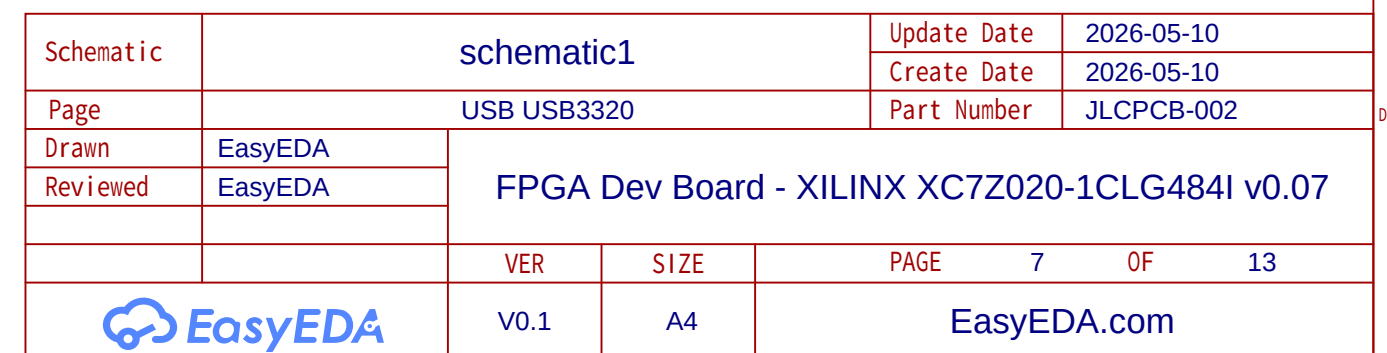
ADBUS7 (SRST_B) is an output that connects to the PS_SRST_B signal on Zynq parts. Like POR_B, this must be buffered with an open-collector output. See <https://etherealwake.com/2024/06/xilinx-ftdi-jtag/>.

From @asmi06: add a comparator (I use MCP6541T-E/OT) to pin ADBUS4 such that if your board is powered externally (and not through USB - which is how most of my boards are powered because USB just doesn't provide enough juice for devices like Artix-A100T), ADBUS4 would only go high if main power is connected

-added 12K pullups on CLK and CS

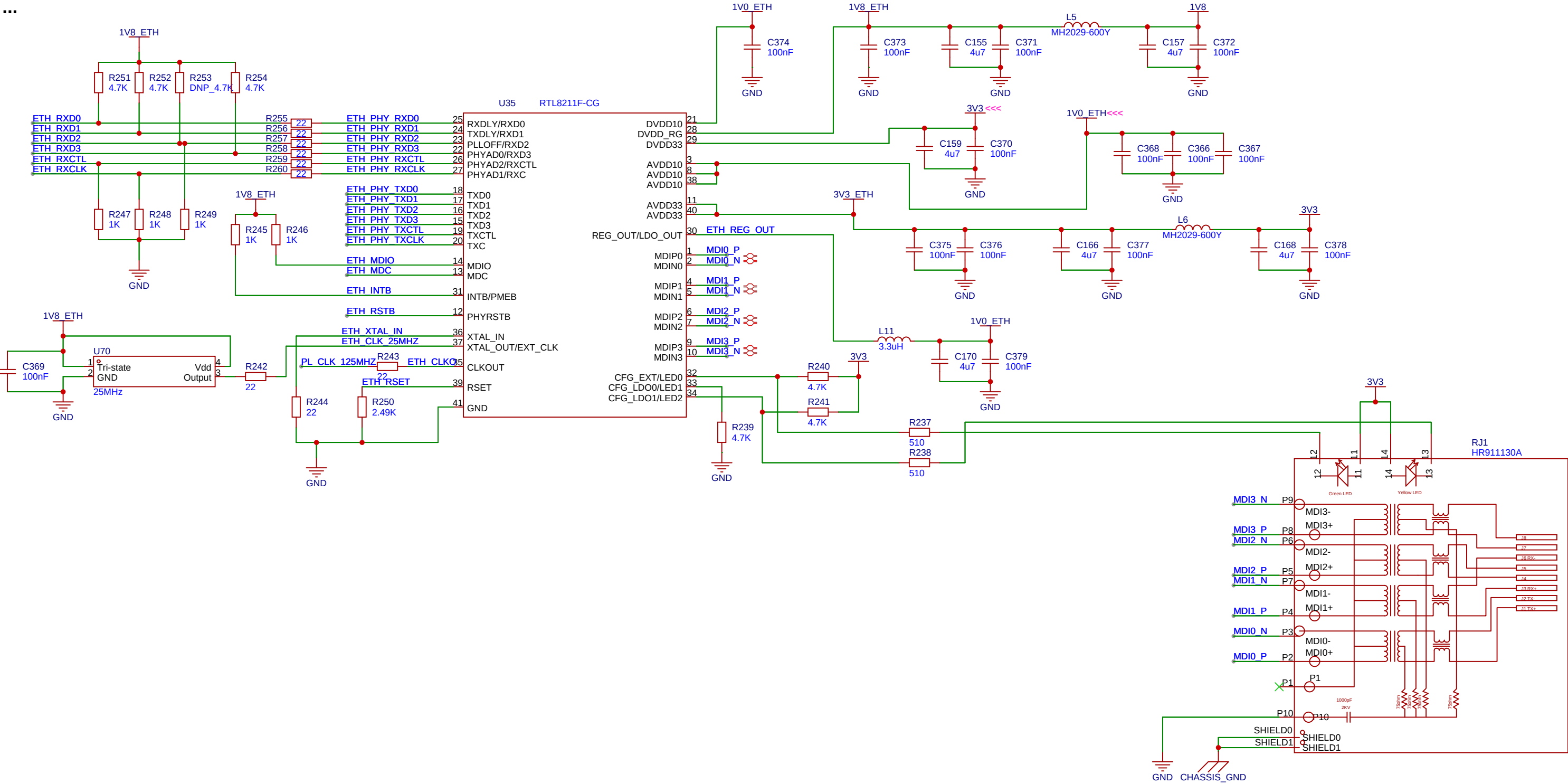
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				Create Date	2026-05-10		
Page	USB FT2232			Part Number	JLCPCB-002		
Drawn	EasyEDA	FPGA Dev Board - XILINX XC7Z020-1CLG484I v0.07					
Reviewed	EasyEDA						
		VER	SIZE	PAGE	6	OF	13
		V0.1	A4	EasyEDA.com			

USB 2.0 ULPI PHY



Ethernet

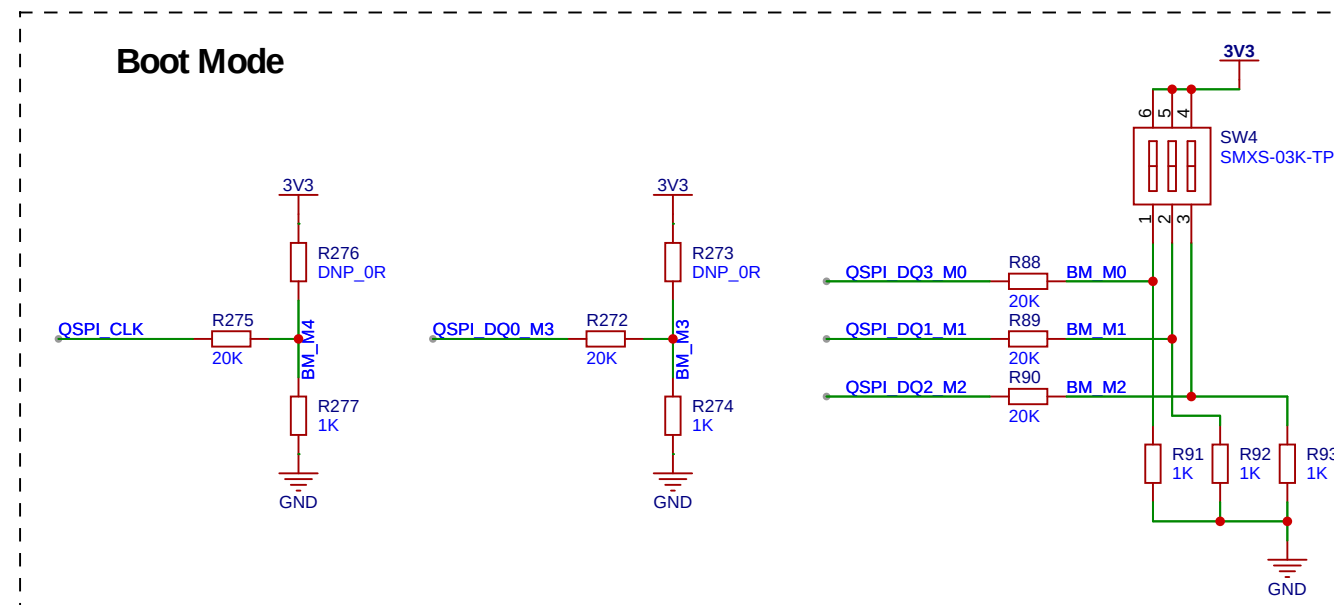
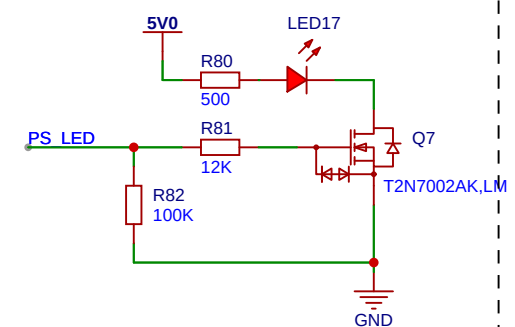
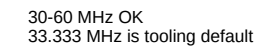
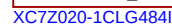
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Reviewed	EasyEDA			
		VER	SIZE	PAGE 8 OF 13
EasyEDA		V0.1	A4	EasyEDA.com

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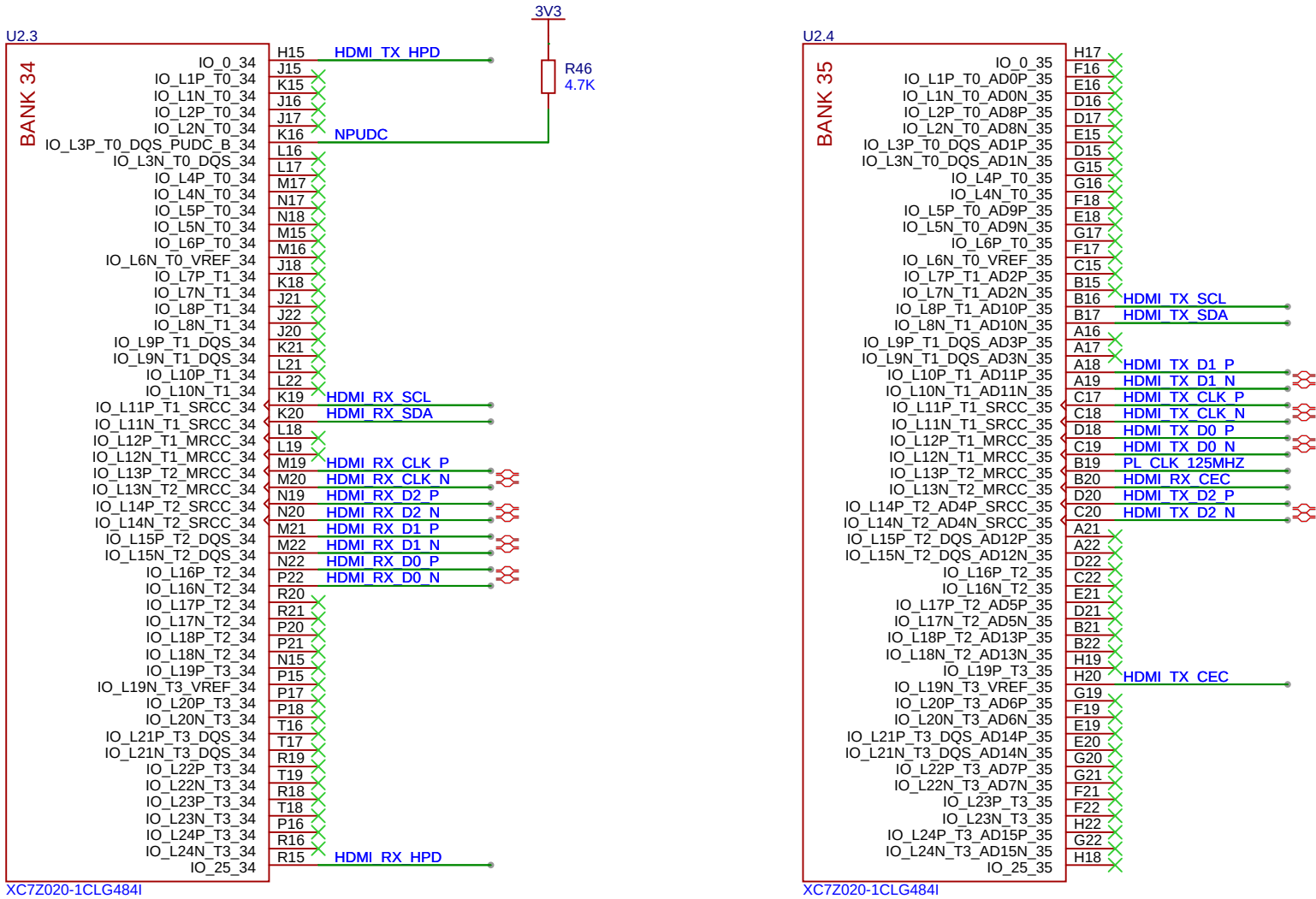
Configure PS pinouts Vivado!



Schematic	schematic1			Update Date	2026-05-10
Page	Processor			Create Date	2026-05-10
Drawn	EasyEDA	FPGA Dev Board - XILINX XC7Z020-1CLG484I v0.07			
Reviewed	EasyEDA				
		VER	SIZE	PAGE	9 OF 13
		V0.1	A4	EasyEDA.com	

Banks 34,35 - HDMI

...

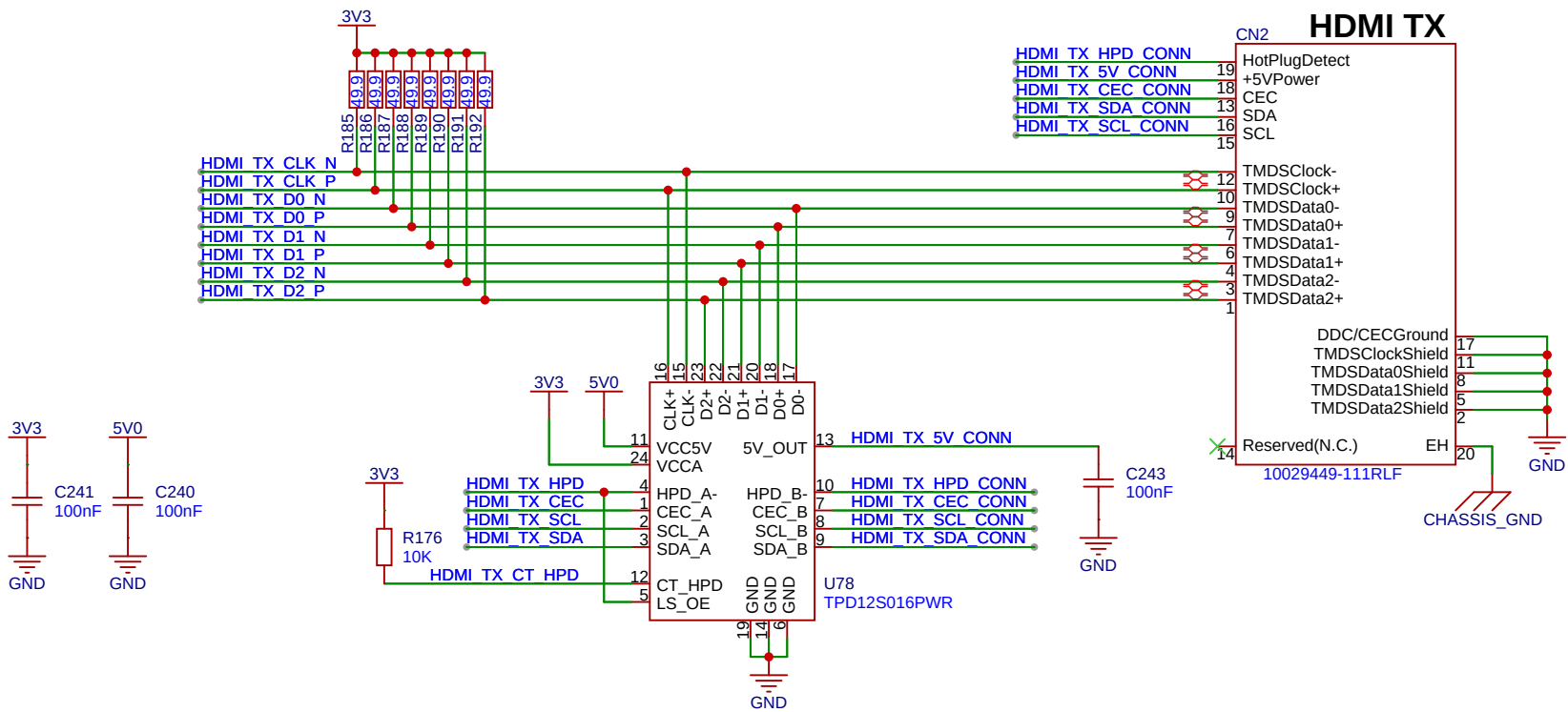


Schematic	schematic1			Update Date	2026-05-10		
				Create Date	2026-05-10		
Page	Banks 34,35 - HDMI, unused			Part Number	JLCPCB-002		
Drawn	EasyEDA	FPGA Dev Board - XILINX XC7Z020-1CLG484I v0.07					
Reviewed	EasyEDA						
		VER	SIZE	PAGE	10	OF	13
		V0.1	A4	EasyEDA.com			

HDMI

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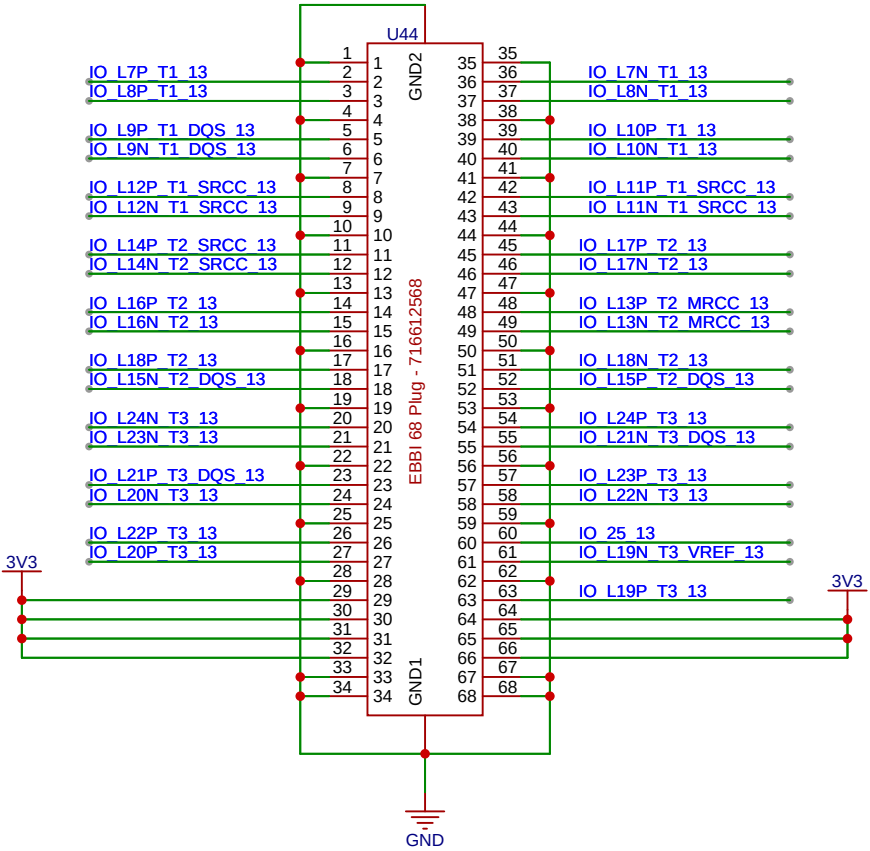
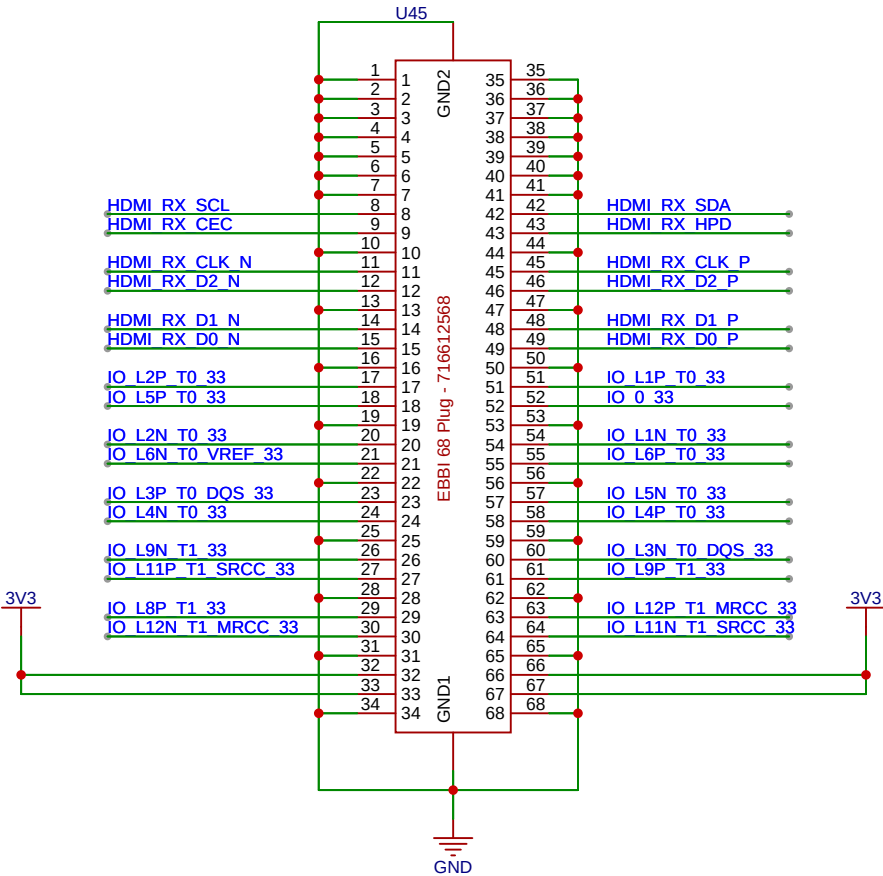
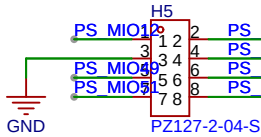
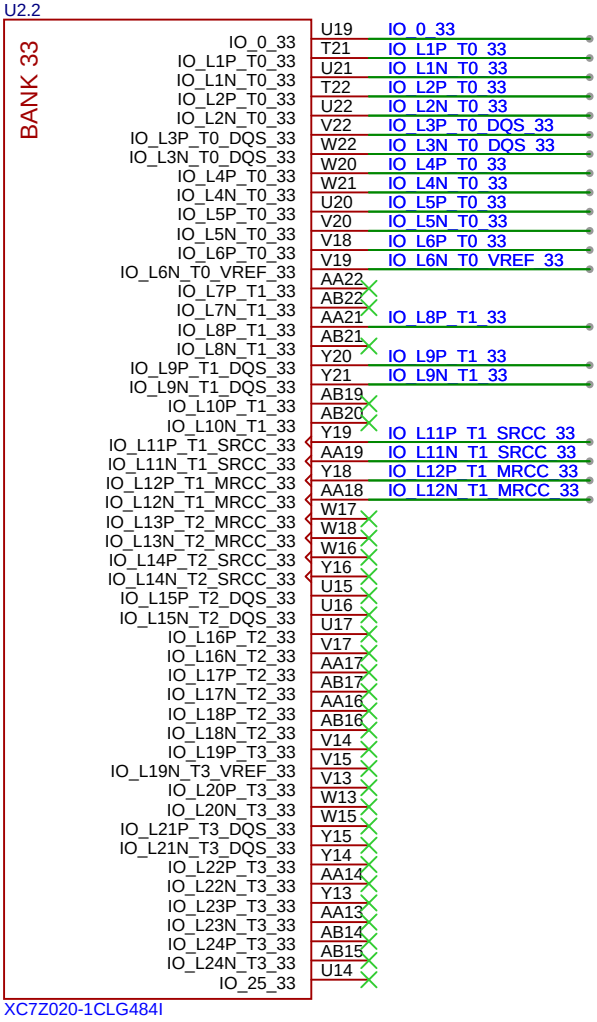
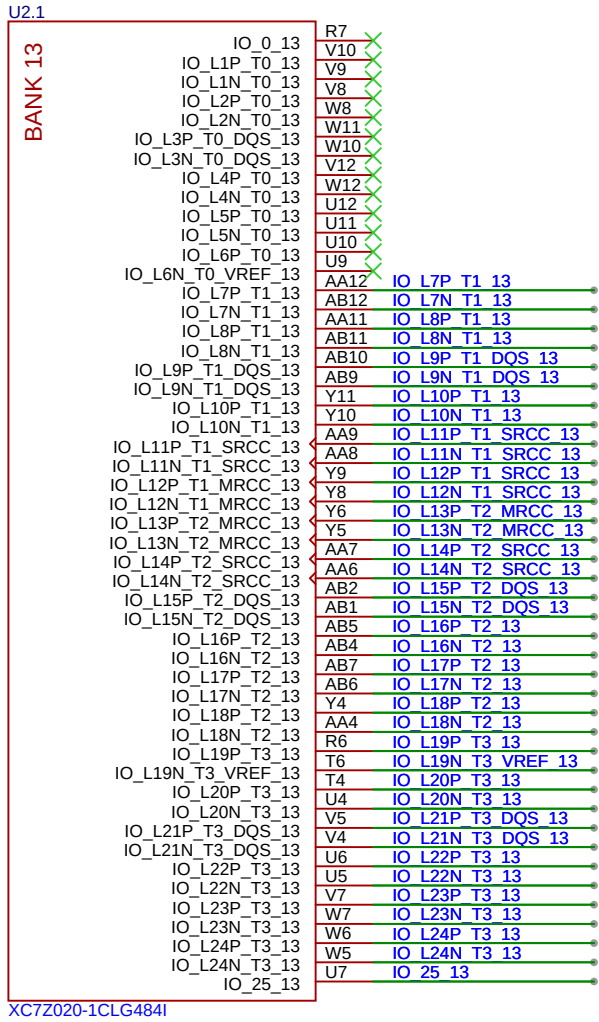
Fly-by routing



Schematic	schematic1			Update Date	2026-05-10		
				Create Date	2026-05-10		
Page	HDMI			Part Number	JLCPCB-002		
Drawn	EasyEDA	FPGA Dev Board - XILINX XC7Z020-1CLG484I v0.07					
Reviewed	EasyEDA						
		VER	SIZE	PAGE	11	OF	13
		V0.1	A4	EasyEDA.com			

Expansion Ports

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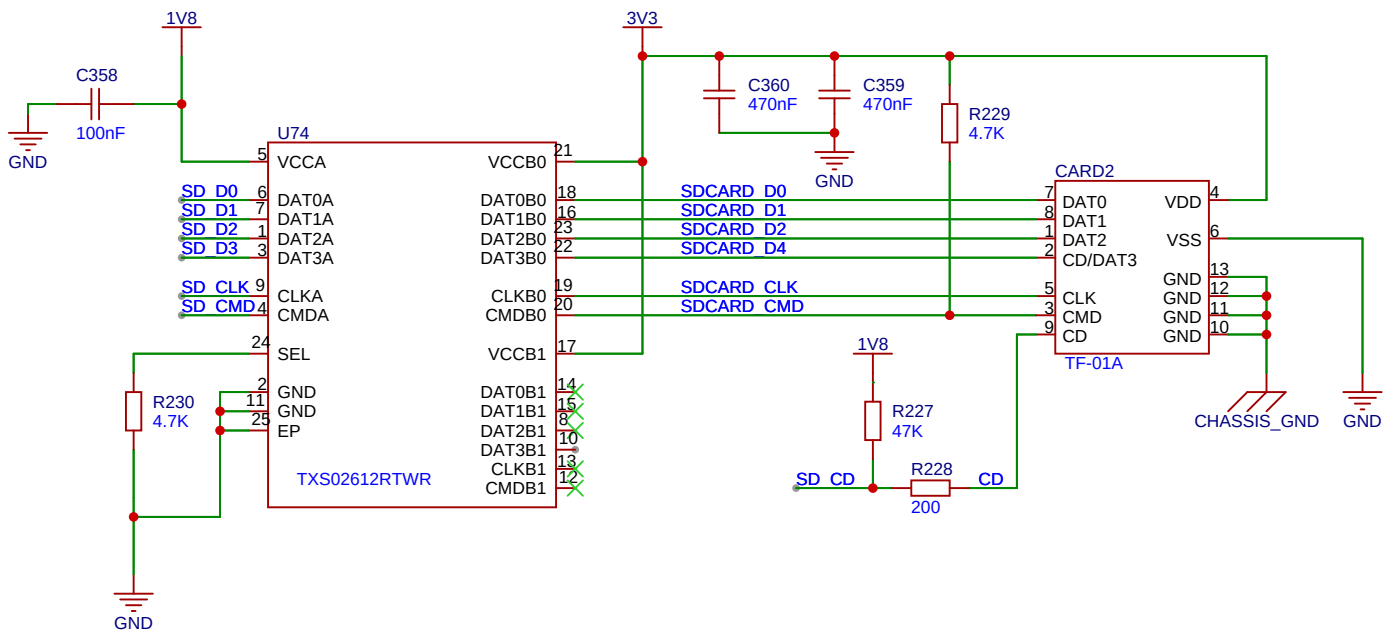


Schematic	schematic1			Update Date	2026-05-10		
				Create Date	2026-05-10		
Page	Banks 13,33 - Expansion			Part Number	JLCPCB-002		
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Reviewed	EasyEDA						
		VER	SIZE	PAGE	12	OF	13
		V0.1	A4	EasyEDA.com			

SD Card Reader

Placeholder...

To do:
-



Schematic	schematic1		Update Date	2026-05-10
			Create Date	2026-05-10
Page	SD Card		Part Number	JLCPCB-002
Drawn	EasyEDA	FPGA Dev Board - XILINX XC7Z020-1CLG484I v0.07		
Reviewed	EasyEDA			
		VER	SIZE	PAGE 13 OF 13
EasyEDA		V0.1	A4	EasyEDA.com